Chinese ADR random-forest long-short

9 October 2026

Part I describes the model and the rule that turns its forecasts into positions; Part II gives the backtest statistics; the appendix lists the hyperparameters chosen each year and the feature importances.

Part I. Model description

What it is

An application of a cross-sectional forecast of next-month stock returns from "Empirical Asset Pricing via Machine Learning" (Gu, Kelly, and Xiu, 2020), applied to the US tradeable Chinese market. This strategy has exploitable alpha primarily due to high borrow costs of rebalancing so many short positions monthly forcing a capital constraint of sub $1M. Further evidence for this being the case is that the same exact strategy run over US equities returns a Sharpe of 0.13, demonstrating the decay of alpha within the US domestic market, as this basic mechanism is undoubtedly used to better effect domestically by various long-short funds, which cannot enter our market due to the liquidity constraint being too tight. The forecasts are turned into an equal-weight long-short portfolio: long the top fifth of the liquid universe, short the bottom fifth of the names that trade at least $3m a day. Nothing in the model uses fundamentals, analysts, news or flows.

Headline result: on a $500,000 account with a 10% volatility target, after commissions, borrow and cash interest, the strategy grows to $1.81m over January 2010 to August 2026, an 8.0% compound annual return at 11.1% volatility, Sharpe 0.75 on daily returns (1.01 monthly), worst drawdown −15.0%, with the money arriving in bursts (2010–11, 2016, 2020–22, 2024–26) around flat years. Before sizing, the dollar-neutral portfolio earns 45% a year gross, Sharpe 1.34, of which the short leg contributes −29% a year and the long leg +17%; it is beta-neutral as well, with a beta of 0.00 and a correlation of 0.00 to the PGJ China ETF (long leg beta 0.95, short leg 0.95).

Universe

US-listed Chinese companies (ADRs and direct listings) assembled from the USCC lists of 2019, 2020 and 2025 and the Nasdaq screener: 453 names over 2000–2026, 302 still listed. Prices are dividend- and split-adjusted daily closes with volume. Twelve SOEs that delisted in 2021–23 are carried to their delisting dates through their Hong Kong lines.

Inputs

Daily prices are cleaned first: isolated prints that revert the next day are dropped, both days of a crash-then-rebound pair are removed, and any remaining single day above +300% (an unadjusted split) is removed. Then at each month-end t, from that name's own history up to t:

Characteristic	Definition
ret_1m	return over month t
mom_6_1, mom_12_1	cumulative return over months t−5..t−1 and t−11..t−1
mom_36_13	cumulative return over months t−35..t−12
vol_1m, vol_3m	standard deviation of daily returns in month t; average of the last three months'
ivol_12m, beta_12m	residual volatility and beta against the equal-weight universe over the trailing 12 months of daily returns
maxret_1m, minret_1m	largest and smallest daily return in month t
dolvol_1m	log of average daily dollar volume in month t
dolvol_trend	dolvol_1m minus its trailing 12-month mean

Characteristic	Definition
illiq_1m	log Amihud ratio: mean of abs(return) / dollar volume, × 1e6
zero_1m	share of zero-return days in month t
high52	month-end price divided by the 52-week high
age	log of months since first price

Each characteristic is converted to its percentile rank within that month's tradeable cross-section (price $\geq$ \$1, $\geq$ \$100k a day) and mapped to [−1, 1]; a missing value becomes 0, the median. The model sees only these sixteen ranks. Ranks must be computed over the same kind of universe the model was fitted on: feeding it ranks from a different universe (say, all US stocks) changes what "+1 on dolvol_1m" means.

Target

The next calendar month's return minus the 1-month T-bill. For fitting only, the target is winsorised at the training sample's 1st and 99th percentiles (about −0.45 and +0.71 for the latest model).

Estimator and refit protocol

A random forest of 300 trees, each split drawing on half of the characteristics, with maximum tree depth in {2, 4, 6} and minimum leaf size in {20, 50} chosen by validation mean squared error. The protocol is an annual refit: in December before test year Y, train on 2001..Y−3, validate on Y−2..Y−1, pick the six-candidate winner, and use it unchanged for the twelve months of Y. No test-year data touches any fit. The hyperparameters chosen each year are listed in Appendix A: shallow trees (depth 2) in most years to 2019 and again for 2026, depth 4 to 6 in 2016–17 and 2020–25.

What the forest has learned

Impurity-based importance for the 2026 forest (Appendix B): zero_1m 0.33, high52 0.23, ivol_12m 0.08, mom_6_1 0.05, dolvol_trend 0.05, minret_1m 0.05, vol_3m 0.05, ret_1m 0.04, everything else below 0.04. The first thing the forest does is separate names with many zero-return days (thin, stale, often halted or about to be) from the rest, then it reads price relative to the 52-week high, idiosyncratic volatility and recent returns. The linear models fitted alongside it in the research agree on the direction of the stable terms: liquid, calm names near their highs score high; thin, volatile, recently-collapsed names score low. In the quintile sorts the short leg carries the result: the bottom quintile of liquid names lost about 27% a year over 2010–26, the top quintile earned about 14% a year, roughly what a volume-weighted index of the universe earned.

Forecast quality out of sample, 2010–26, all tradeable names: R^2 against a zero forecast 0.35%, rank IC 0.080 (t 7.0), 0.081 on the liquid names (t 6.1).

From forecasts to positions

At each month-end, among tradeable names with $\geq$ \$1m a day (25 or more required, else flat): long the top quintile by forecast, equal weight. Among those also trading $\geq$ \$3m a day (10 or more required, else flat): short the bottom quintile, equal weight, at least five names. Hold one calendar month, buy-and-hold inside it, re-form at the next month-end. The \$3m floor is applied only here; the model is fitted on the whole tradeable panel. Its purpose is borrow: names under \$3m a day cost more to borrow at retail than they earn, and removing them raises the gross spread as well (Sharpe 1.37 against 1.19 unrestricted) because they are the noisiest shorts. Raising the floor further costs more than it saves: at \$10m the gross Sharpe is 1.04 and at \$25m 0.50, as the names that lose the most are no longer in the pool. The \$3m rule carries an average borrow of about 7% a year on the short leg (14% under the harsh schedule), against 2% at \$10m, and is the best net of borrow under both schedules.

Sizing in the \$500k backtest: dollars a side = min(80%, 10% ÷ σ) × equity, resized monthly, σ = 0.7 × trailing 60-day realised volatility of the one-dollar-a-side long-short + 0.3 × its expanding mean. Fixed exposure (50% a side) earns about twice as much at one and a half times the drawdown and the same Sharpe; the volatility target cut exposure to 12–25% a side in 2025–26 when the unit volatility ran at 40–70%. The sizing table in the statistics compares the alternatives.

Part II. Backtest statistics

All numbers are out of sample in the sense that no model saw data from its test year; the design (features, universe, model menu, the $3m floor) was fixed with the research and its literature in view. Window January 2010 to August 2026.

1. Forecast quality

All tradeable names, 2010–26.

	R^2 vs zero forecast	Rank IC (t)	Rank IC, liquid names (t)
Random forest	0.35%	0.080 (7.0)	0.081 (6.1)
Ensemble of five models, for reference	0.31%	0.087 (8.2)	0.089 (7.0)
Huber OLS, for reference	0.44%	0.115 (9.7)	0.107 (7.3)

2. Quintile portfolios, unrestricted short leg

Liquid universe ($\geq$ $1m a day), equal weight, 191 months, 62 names a month.

	Random forest
Long-short return, annualised	+41.1%
Long-short Sharpe, gross / net of 20 bp a side and 3% borrow	1.16 / 0.93 (t 4.6)
Long-short worst drawdown	−37%
Top quintile / bottom quintile excess return	+13.9% / −27.2% a year
Equal-weight universe excess return	−1.5% a year
Top quintile minus equal-weight universe, Sharpe	0.89
Turnover, long / short leg	48% / 58% a month
Sub-period Sharpe 2010–15 / 2016–20 / 2021–26	1.31 / 0.99 / 1.19
Calendar years, long-short gross (%)	2010 +33, 2011 +73, 2012 −4, 2013 +28, 2014 +36, 2015 +57, 2016 +53, 2017 +7, 2018 −17, 2019 −7, 2020 +226, 2021 +48, 2022 +23, 2023 +44, 2024 +54, 2025 +30, 2026 (to August) +82

3. Borrow at retail scale and the $3m short-leg rule

Schedules by daily dollar volume, assumed not quoted: base 1% above $25m, 4% at $10–25m, 12% at $3–10m, 35% below $3m; harsh 2 / 8 / 25 / 60%.

Short-leg rule	Gross Sharpe	Average borrow	Net Sharpe (base / harsh)	Net 2021–26	Shorts a month
All liquid names, flat 3% borrow	1.19	3%	0.96	1.00	12
All liquid names, tiered	1.19	16 / 29%	0.60 / 0.24	0.68 / 0.36	12
Names above $3m a day (exported rule)	**1.37**	**7 / 14%**	**1.02 / 0.81**	**1.18 / 1.00**	**9**
Names above $10m a day	1.04	2 / 5%	0.84 / 0.77	0.91 / 0.84	7
Names above $25m a day	0.50	1 / 2%	0.35 / 0.33	0.39 / 0.36	6
Long leg, short PGJ at 1%	0.53	1%	0.36	−0.07	

4. The $500,000 account, 2010–26

Daily P&L on the monthly holdings; dollars a side = min(80%, 10% $\div$ σ) $\times$ equity with σ from the unit long-short's 60-day volatility blended with its long-run mean; 20 bp a side; borrow per name by volume; free cash at the bill less 0.5%, short proceeds above $100k at the bill less 1.25%; Reg T margin.

	Random forest (headline)	Same account on the ensemble forecasts
Final equity from $500,000	**$1.81m**	$1.36m
CAGR / volatility	**8.0% / 11.1%**	6.2% / 10.9%
Sharpe, daily / monthly	**0.75 / 1.01**	0.60 / 0.81
Worst drawdown	−15.0% (Oct 2017 to Mar 2019, recovered Mar 2020)	−12.2%
Worst month / hit rate	−4.6% / 58%	−5.0% / 56%
Sub-period Sharpe 2010–15 / 2016–20 / 2021–26	0.55 / 0.76 / 0.99	0.32 / 0.64 / 0.92
Exposure a side, average (range)	24% (5 to 37%)	24%
Margin pledged, average / peak (Reg T initial); maintenance peak; call days	24% / 41%; 22%; 0	
Commissions and spread / borrow (6.6% a year) / cash interest earned	$200k / $232k / +$209k	
Flat months (too few names)	10 (Dec 2009, Jul 2010, Dec 2011, May 2012, Jun 2012, Jul 2012, Aug 2012, Sep 2012, Oct 2012, Nov 2012)	

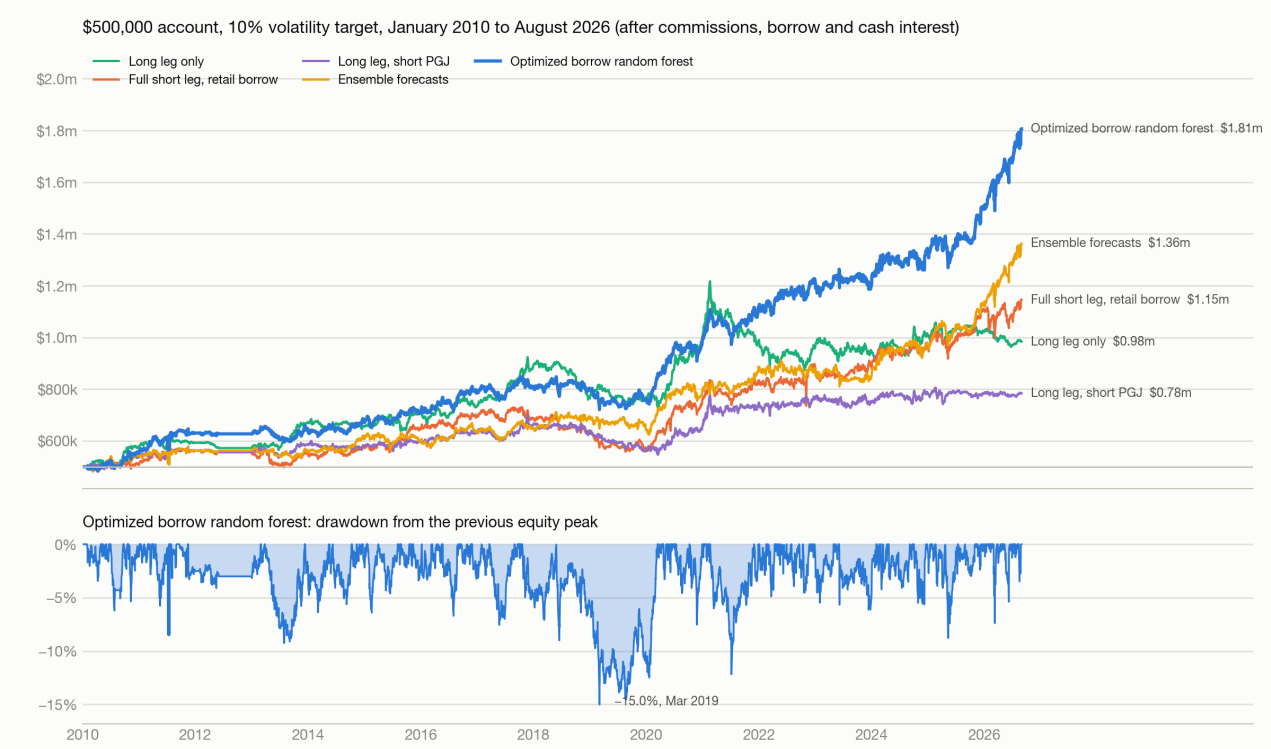


Figure 2. Account equity and drawdown, January 2010 to August 2026: the optimized-borrow random forest against the same account on the ensemble forecasts, the full short leg at retail borrow, the long leg alone and the long leg hedged with PGJ.

Calendar years, return on the account (%): 2010 +10.6, 2011 +14.2, 2012 −0.4, 2013 +3.0, 2014 +4.5, 2015 +5.2, 2016 +12.0, 2017 +1.8, 2018 −1.0, 2019 −5.5, 2020 +31.3, 2021 +10.6, 2022 +11.4, 2023 −2.5, 2024 +12.0, 2025 +12.7, 2026 (to August) +19.8. The best five months are 25% of the P&L, the best ten 47%.

Sizing (the same holdings, different exposure rules; all after costs, borrow and cash interest):

Sizing	CAGR	Final equity	Volatility	Sharpe daily / monthly	Worst drawdown	Margin peak (Reg T initial)
10% volatility target (headline)	8.0%	$1.81m	11.1%	0.75 / 1.01	−15%	41%
15% volatility target	11.4%	$3.02m	16.7%	0.73 / 0.96	−23%	62%
20% volatility target	14.7%	$4.91m	22.4%	0.72 / 0.94	−29%	85%
Fixed 50% a side	16.5%	$6.45m	24.8%	0.74 / 1.02	−23%	73%
Fixed 80% a side	25.3%	$21.61m	40.6%	0.76 / 0.99	−36%	130%

Cost and borrow variants (Sharpe daily): harsh borrow 0.60; flat 8% borrow 0.72; no borrow 0.89; 10 bp costs

0.81; 40 bp costs 0.63; 60-day volatility without the long-run blend 0.72. Short-leg floor at $10m instead of $3m: $1.54m, CAGR 7.0%, Sharpe 0.75 / 0.85, drawdown −19%; at $25m: $863k, 0.40.

5. Monte Carlo and nulls

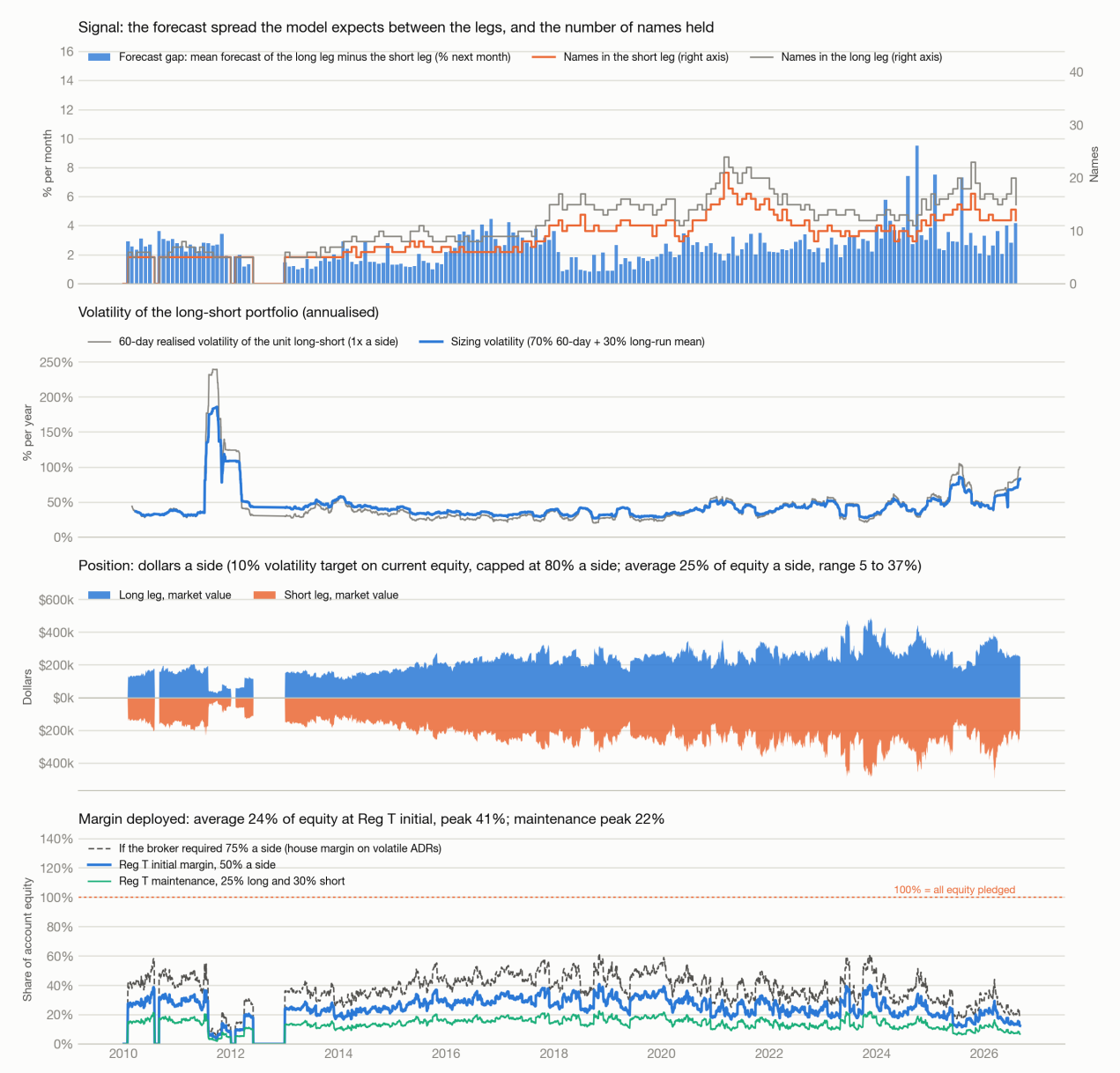


Figure 3. Signal (forecast gap and names held), volatility of the long-short, dollars a side, and margin deployed, 2010 to 2026.

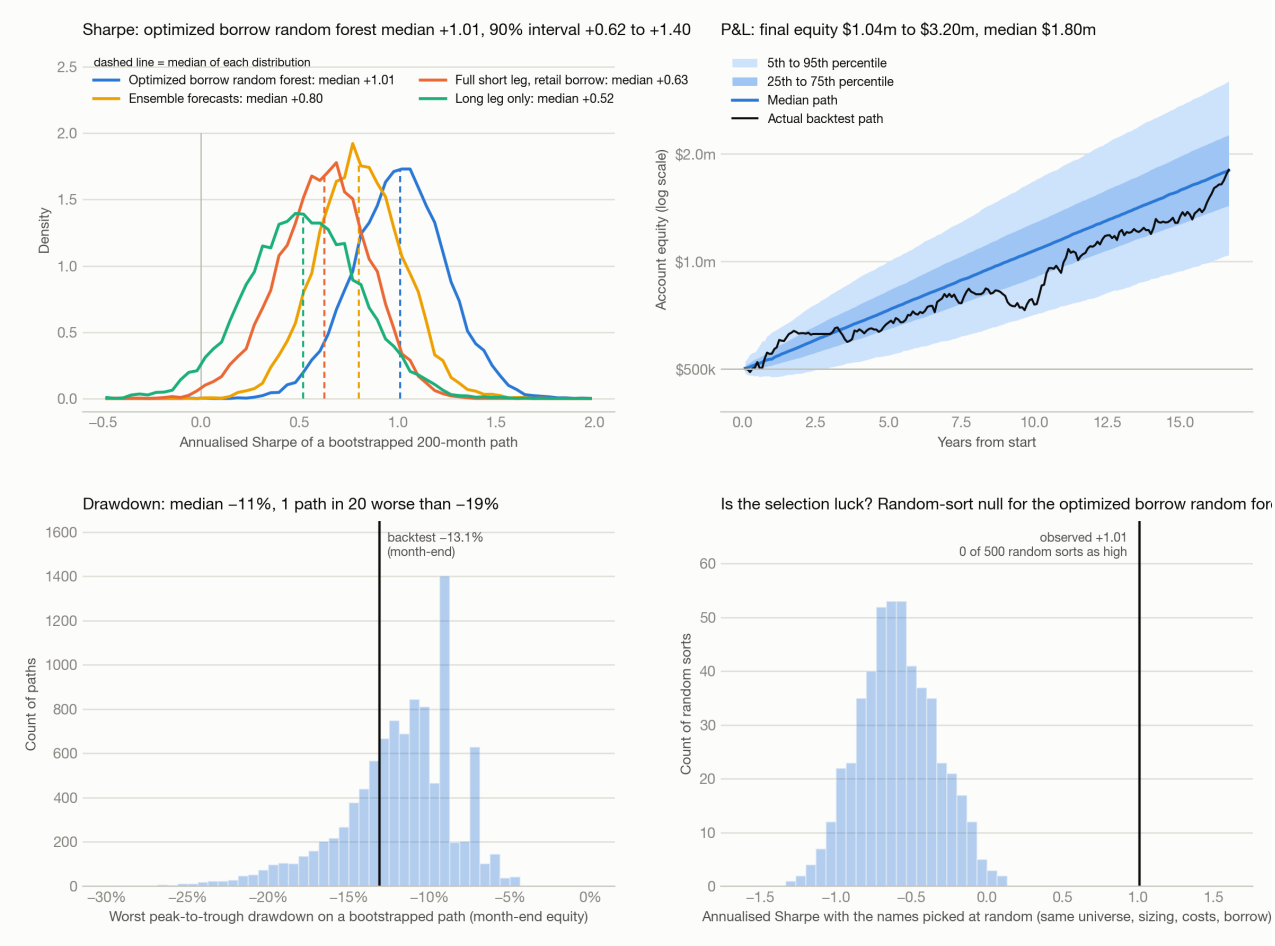


Figure 4. Bootstrap Sharpe densities, equity fan against the actual path, drawdown distribution, and the random-sort null.

6. The last five years as a fresh account

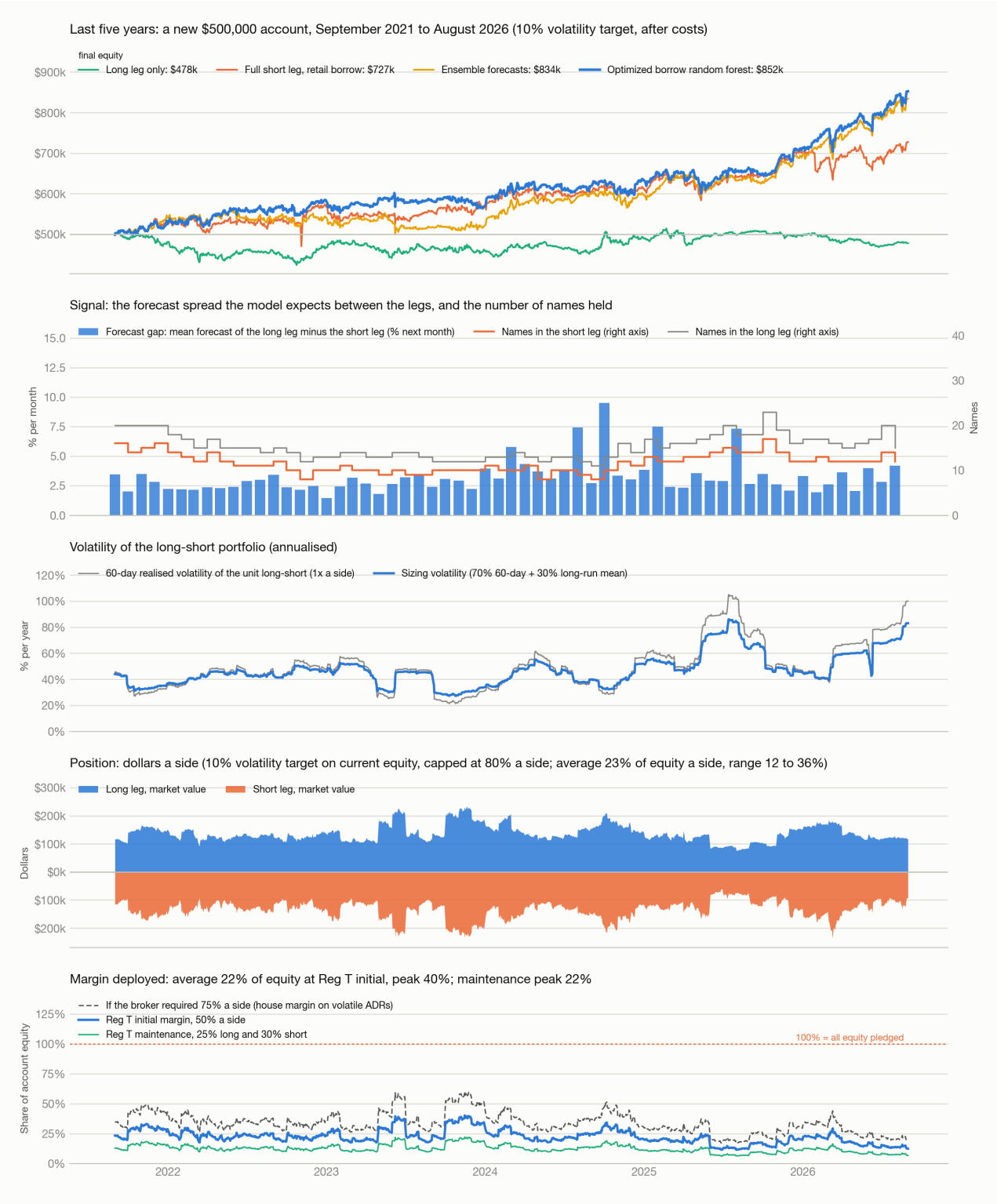


Figure 5. A fresh account from September 2021: equity, signal, volatility, dollars a side, margin.

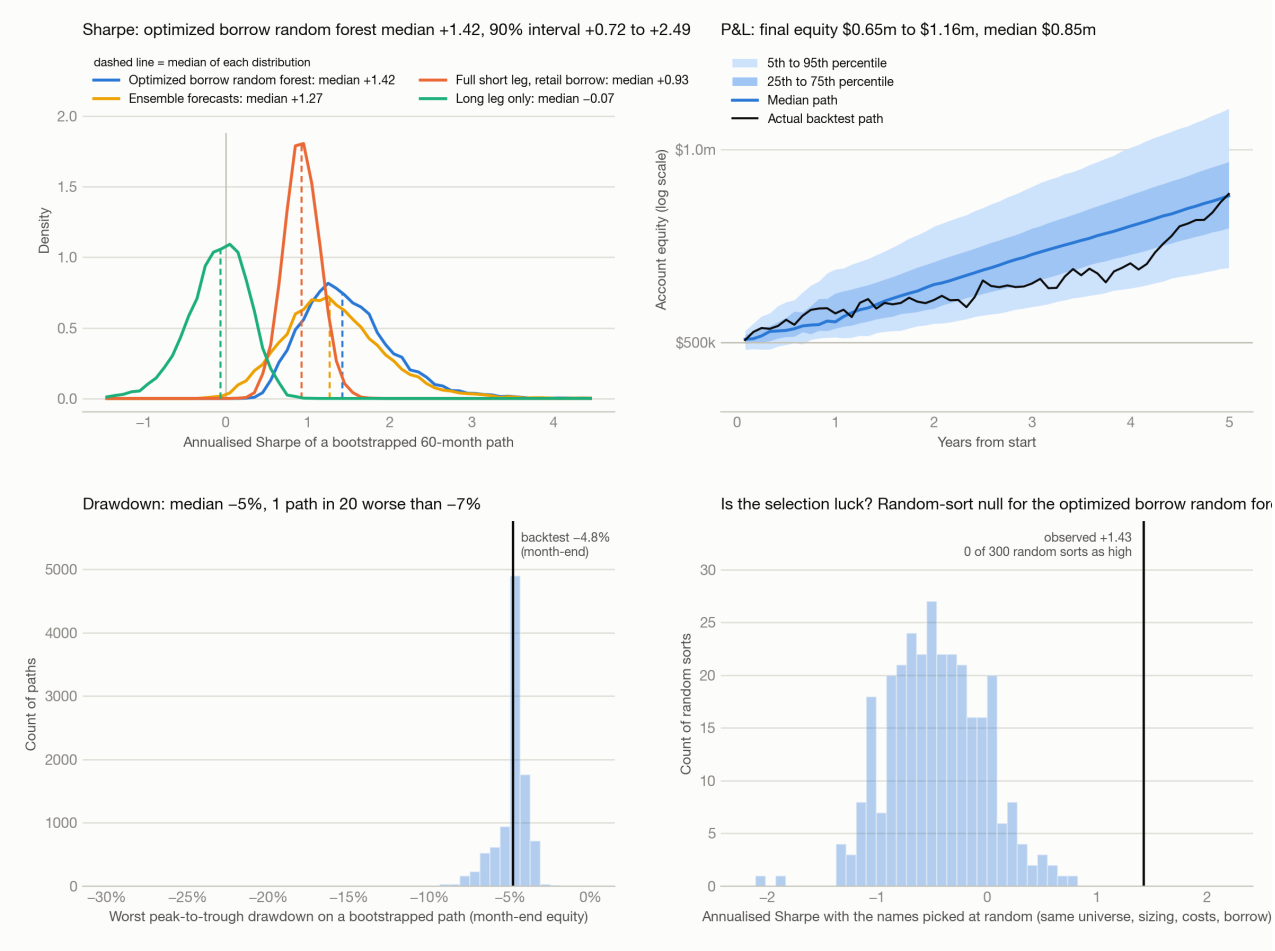


Figure 6. Monte Carlo on the 60 months.

7. Cross-checks and limits

- Price data: Yahoo and an independent archive agree on 99.95% of overlapping stock-months. Two data bugs were found and fixed in review (an asymmetric return filter that turned bad prints into −100% months, and a label-indexing error in the cleaning step that nulled unrelated days when sources were concatenated); the numbers here are from the corrected pipeline.
- Survivorship: names that delisted after April 2020, other than twelve SOEs carried through Hong Kong, are missing for their final months. The volume-weighted universe correlates 0.92 with PGJ and trails it by 4% a year in 2021–26, so the survivor premium is not driving the recent numbers; the small-cap delistings sit in the short leg and remain unmeasured.
- Concentration: the pool above $3m a day was 14 to 58 names a month until 2017 and 43 to 108 since, so the short leg has held 5 to 11 names before 2018 and 8 to 21 since; at the minimum of five names a single short is 20% of the leg.

Appendix

A. Hyperparameters chosen by validation, by test year

Each December before test year Y the forest is trained on 2001..Y−3 and the six candidates (max_depth 2, 4, 6 × min_samples_leaf 20, 50) are scored on Y−2..Y−1. Refitting with these settings reproduces the saved forecasts to machine precision.

Test year	Training rows (names)	Validation rows	max_depth	min_samples_leaf	Validation MSE
2010	1,657 (41)	907	2	20	0.0333
2011	2,097 (47)	1,065	2	50	0.0252
2012	2,564 (54)	1,270	2	50	0.0254

Test year	Training rows (names)	Validation rows	max_depth	min_samples_leaf	Validation MSE
2013	3,162 (68)	1,271	2	20	0.0250
2014	3,834 (73)	1,300	2	50	0.0235
2015	4,433 (77)	1,563	2	20	0.0237
2016	5,134 (80)	1,711	6	50	0.0236
2017	5,996 (90)	1,757	6	50	0.0209
2018	6,845 (100)	1,971	2	20	0.0216
2019	7,753 (115)	2,468	2	50	0.0283
2020	8,816 (138)	2,964	4	50	0.0300
2021	10,221 (173)	2,918	4	50	0.0374
2022	11,780 (201)	3,105	4	50	0.0427
2023	13,139 (226)	3,166	6	50	0.0457
2024	14,885 (248)	2,722	6	20	0.0461
2025	16,305 (256)	2,754	4	20	0.0441
2026	17,607 (276)	3,315	2	20	0.0488

B. Impurity-based feature importance

Characteristic	rf_2026	rf_next
zero_1m	0.326	0.291
high52	0.231	0.076
ivol_12m	0.082	0.072
mom_6_1	0.051	0.044
dolvol_trend	0.051	0.037
minret_1m	0.050	0.057
vol_3m	0.046	0.061
ret_1m	0.043	0.063
beta_12m	0.027	0.050
maxret_1m	0.024	0.041
mom_12_1	0.022	0.033
dolvol_1m	0.015	0.042
age	0.011	0.040
illiq_1m	0.010	0.034
vol_1m	0.008	0.038
mom_36_13	0.003	0.021